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GATE CSE · DATABASE MANAGEMENT SYSTEM · CHAPTER-WISE PYQS

Er Diagrams

GATE CSE Database Management System · Previous Year Questions with Answer Key

16	13	2004 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	10 questions
Marks 2	6 questions
Answer key	at the end

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Marks 1 10 questions

Q1. GATE CSE 2024 Set 2

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

(A) The weak entity set MUST have total participation in the identifying relationship

(B) The owner entity set MUST have total participation in the identifying relationship

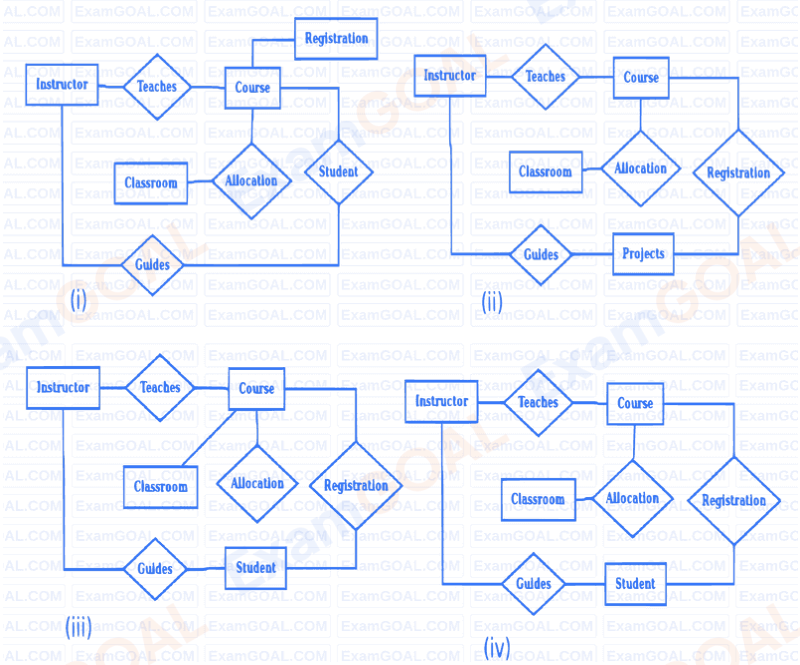
(C) Both weak and owner entity sets MUST have total participation in the identifying relationship

(D) Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

Open solution

Q2. GATE CSE 2024 Set 1

Let S be the specification : "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



(A) (i) (B) (ii) (C) (iii) (D) (iv)

Open solution

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Q3. GATE CSE 2023

Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?

(A) Number of attributes of its relation schema.

(B) Number of tuples stored in the relation.

(C) Number of entries in the relation.

(D) Number of distinct domains of its relation schema.

Open solution

Q4. GATE CSE 2020

Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

(A) Rectangles with double/bold border

(B) Diamonds with double/bold border

(C) Ovals that contain underlined identifiers

(D) Ovals with double/bold border

Open solution

Q5. GATE CSE 2016 Set 1

Which of the following is **NOT** a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY?

(A) $VXYZ$ (B) $VWXZ$ (C) $VWXY$ (D) $VWXYZ$

Open solution

Q6. GATE CSE 2014 Set 2

Given an instance of the **STUDENTS** relation as shown below:

StudentID	*StudentName*	*StudentEmail*	*StudentAge*	*CPI*
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

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For **(StudentName, StudentAge)** to be a key for this instance, the value X should **NOT** be equal to _____.

Open solution

Q7. GATE CSE 2013

An index is clustered, if

(A) It is on a set of fields that form a candidate key.

(B) It is on a set of fields that include the primary key.

(C) The data records of the file are organized in the same order as the data entries of the index.

(D) The data records of the file are organized not in the same order as the data entries of the index.

Open solution

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Q8. GATE CSE 2012

Open solution

Given the basic ER and relational models, which of the following is INCORRECT?

(A) An attribute of an entity can have more than one value

(B) An attribute of an entity can be composite

(C) In a row of a relational table, an attribute can have more than one value

(D) In a row of a relational table, an attribute can have exactly one value or a NULL value

Q9. GATE CSE 2011

Open solution

Consider a relation table with a single record for each registered student with the following attributes.

1. $Registration\ Num$: Unique registration number of each registered student.
2. UID : Unique identity number, unique at the national level for each citizen.
3. $BankAccount_Num$: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attribute stores the primary account number.
4. $Name$: Name of the student
5. $Hostel\ Room$: Room number of the hostel.

Which of the following options is INCORRECT?

(A) $BankAccount\ Num$ is candidate key

(B) $Registration\ Num$ can be a primary key

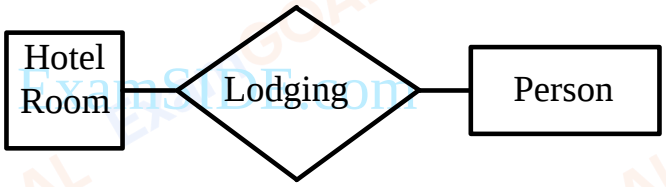
(C) UID is a candidate key if all students are from the same country

(D) If S is a superkey such that $S \cap UID$ is NULL then $S \cup UID$ is also a superkey

Q10. GATE CSE 2005

Open solution

Consider the entities 'hotel room', and 'person' with a many to many relationship 'lodging' as shown below



If we wish to store information about the rent payment to be made by person(s) occupying different hotel rooms, then this information should appear as an attribute of

(A) Person (B) Hotel Room (C) Lodging (D) None of these

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Marks 2 6 questions

Q11. GATE CSE 2018

In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set $E1$ to entity set $E2$. Assume that $E1$ and $E2$ participate totally in R and that the cardinality of $E1$ is greater than the cardinality of $E2$

Which one of the following is true about R?

(A) Every entity in $E1$ is associated with exactly one entity in $E2$.

(B) Some entity in $E1$ is associated with more than one entity in $E2$.

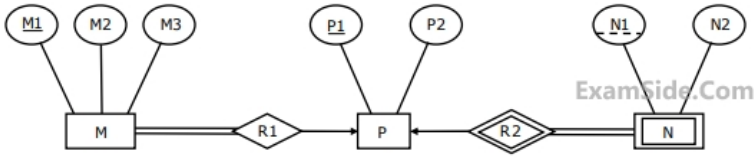
(C) Every entity in $E2$ is associated with exactly one entity in $E1$.

(D) Every entity in $E2$ is associated with at most one entity in $E1$.

Open solution

Q12. GATE CSE 2008

Consider the following ER diagram



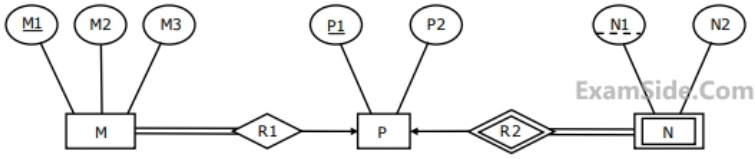
The minimum number of table needed to represent M, N, P, $R1$, $R2$ is

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

Q13. GATE CSE 2008

Consider the following ER diagram



Which of the following is a correct attribute set for one of the tables for the correct answer to the above question?

(A) $\{M1, M2, M3, P1\}$ (B) $\{M1, P1, N1, N2\}$

(C) $\{M1, P1, N1\}$ (D) $\{M1, P1\}$

Open solution

Q14. GATE CSE 2005

Let E_1 and E_2 be two entities in an **E-R** diagram with simple single-valued attributes. R_1 and R_2 are two relationships between E_1 and E_2, where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

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Q15. GATE CSE 2005

Open solution

The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade.

A	B
2	4
3	4
4	3
5	2
7	2
9	5
6	4

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The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(2, 4)$ is deleted is:

(A) $(3, 4)$ and $(6, 4)$

(B) $(5, 2)$ and $(7, 2)$

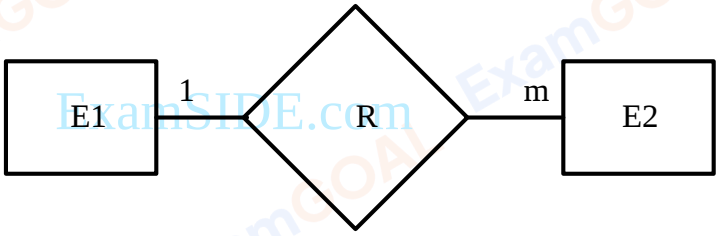
(C) $(5, 2)$, $(7, 2)$ and $(9, 5)$

(D) $(3, 4)$, $(4, 3)$ and $(6, 4)$

Q16. GATE CSE 2004

Open solution

Consider the following entity relationship diagram (ERD), where two entities $E1$ and $E2$ have a relation R of cardinality $1 : m$.



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The attributes of $E1$ are $A11$, $A12$ and $A13$ where $A11$ is the key attribute. The attributes of $E2$ are $A21$, $A22$ and $A23$ where $A21$ is the key attribute and $A23$ is a multi-valued attribute. Relation R does not have any attribute. A relational database containing minimum number of tables with each table satisfying the requirements of the third normal form $(3NF)$ is designed form the above ERD. The minimum number of tables in the database is

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

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Answer Key

Marks 1

1. **A**	2. **D**	3. **A**	4. **B**	5. **B**
6. **19**	7. **C**	8. **C**	9. **A**	10. **C**

Marks 2

11. **A**	12. **B**	13. **A**	14. **B**	15. **C**
16. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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